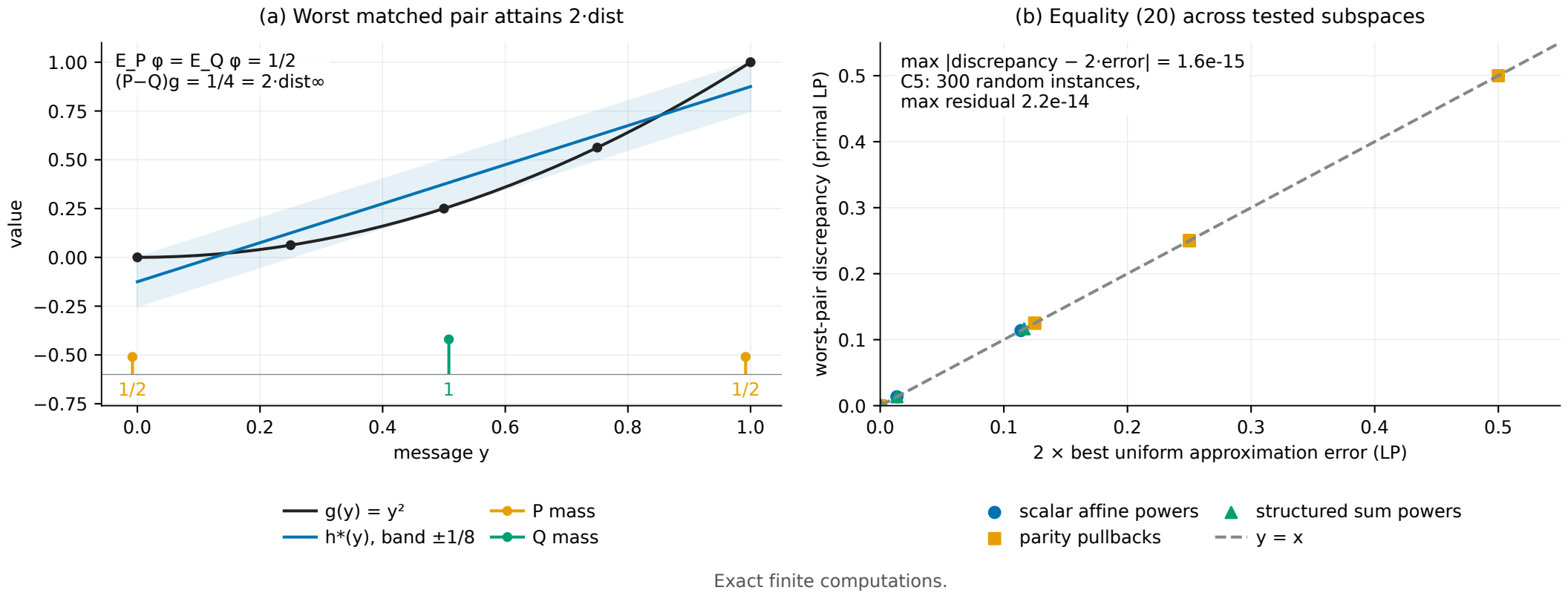
(a) Worst matched pair attains 2·dist
E_P φ = E_Q φ = 1/2
(P−Q)g = 1/4 = 2·dist∞
value
message y
1/2
1
1/2
g(y) = y²
h*(y), band ±1/8
P mass
Q mass
(b) Equality (20) across tested subspaces
max |discrepancy − 2·error| = 1.6e-15
C5: 300 random instances,
max residual 2.2e-14
worst-pair discrepancy (primal LP)
2 × best uniform approximation error (LP)
scalar affine powers
parity pullbacks
structured sum powers
y = x
Exact finite computations.